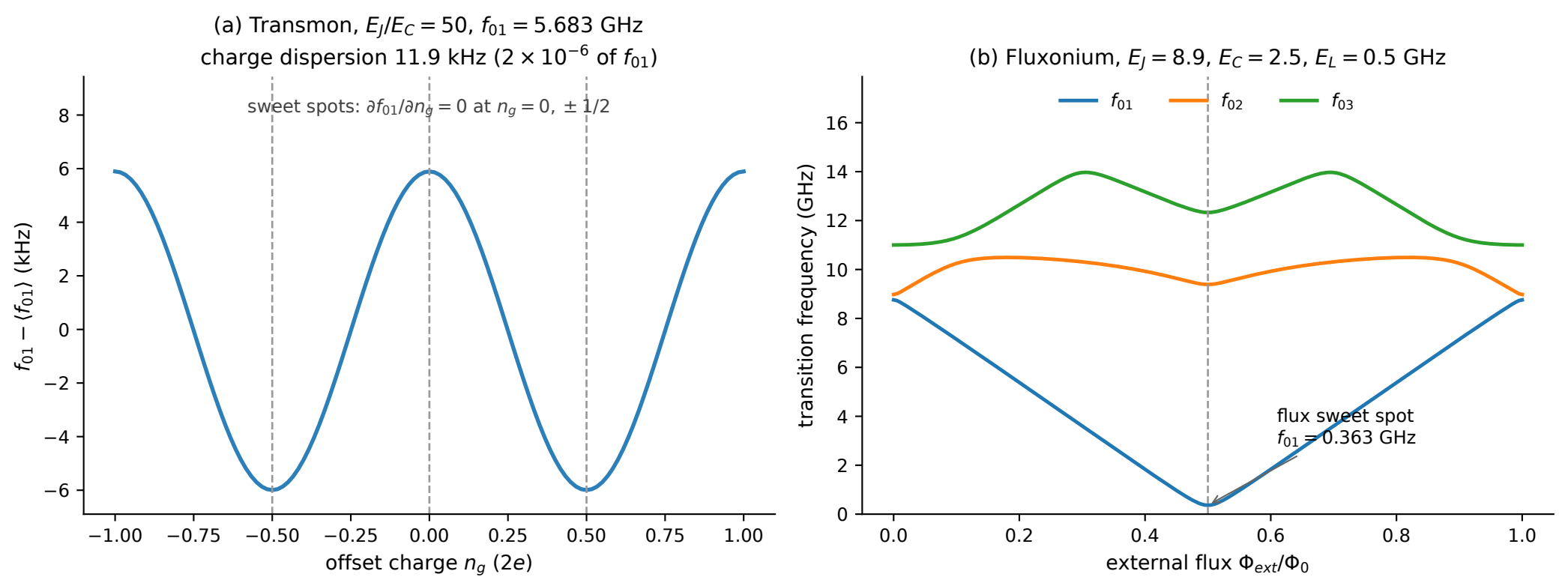

(a) Transmon, $E_J/E_C = 50$, $f_{01} = 5.683$ GHz
charge dispersion 11.9 kHz (2×10^{-6} of f_{01})
sweet spots: $\partial f_{01}/\partial n_g = 0$ at $n_g = 0, \pm 1/2$
$f_{01} - \langle f_{01} \rangle$ (kHz)
offset charge n_g ($2e$)
(b) Fluxonium, $E_J = 8.9$, $E_C = 2.5$, $E_L = 0.5$ GHz
f_{01}
f_{02}
f_{03}
transition frequency (GHz)
flux sweet spot
$f_{01} = 0.363$ GHz
external flux Φ_{ext}/Φ_0